

Senior Data Engineer

Company: Northwind Analytics (fictional) **Location:** Hybrid, Gurugram, India **Employment type:** Full-time **Experience:** 5-8 years

About the Role

Northwind Analytics builds data products for retail and logistics clients. We are hiring a Senior Data Engineer to design and operate the batch and streaming pipelines that feed our analytics platform and ML models. You will own pipelines end to end, from ingestion to serving, and mentor mid-level engineers.

Responsibilities

- Design, build, and maintain scalable batch and streaming data pipelines processing 500M+ events per day.
- Develop and optimize Spark jobs (PySpark) for large-scale transformations, including handling skew, partitioning, and join strategies.
- Orchestrate workflows with Apache Airflow, including dependency management, backfills, retries, and SLAs.
- Model data in a cloud data warehouse (Snowflake or BigQuery) using dimensional and incremental modelling patterns.
- Implement data quality checks, lineage, and monitoring to keep pipelines reliable and auditable.
- Partner with data scientists, analysts, and product managers to translate business needs into data contracts and schemas.
- Lead design reviews and code reviews; mentor junior and mid-level engineers.
- Drive cost optimization for compute and storage across the data platform.
- Participate in on-call rotation and lead incident post-mortems for data outages.

Required Qualifications

- 5+ years of experience in data engineering or backend engineering with a data focus.
- Strong Python skills, including testing, packaging, and writing maintainable production code.
- Advanced SQL: window functions, query plans, performance tuning.
- Hands-on experience with Apache Spark (PySpark) on large datasets.
- Experience with Airflow or a comparable orchestrator.

- Experience with at least one cloud platform (AWS, GCP, or Azure) and its data services.
- Solid understanding of data modelling, ETL/ELT design, and data warehousing concepts.
- Familiarity with Git, CI/CD, and infrastructure-as-code practices.

Preferred Qualifications

- Experience with streaming technologies such as Kafka or Kinesis.
- Experience with dbt and modern ELT workflows.
- Knowledge of lakehouse formats (Delta Lake, Iceberg, or Hudi).
- Exposure to Docker and Kubernetes.
- Experience supporting ML feature pipelines.

Tech Stack

Python, PySpark, SQL, Airflow, Kafka, Snowflake, dbt, AWS (S3, Glue, EMR), Terraform, GitHub Actions, Datadog

What Success Looks Like (First 12 Months)

- Reduce end-to-end latency of the core events pipeline from 4 hours to under 1 hour.
- Cut monthly compute spend on Spark workloads by 20% through job optimization.
- Introduce automated data quality checks covering all tier-1 datasets.
- Mentor two engineers toward mid-to-senior level growth.

Interview Process

1. Recruiter screen (30 min)
2. Technical screen: Python and SQL (60 min)
3. System design: data pipeline architecture (60 min)
4. Behavioral and collaboration interview (45 min)
5. Hiring manager conversation (30 min)